

Constant Acceleration in 1D

Write down s, u, v, a, t . Choose the positive direction. The letter you are not given and not asked for chooses the equation.



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THE TOPIC ON ONE MAP

Constant Acceleration in 1D

01 - The SUVAT toolkit

five equations

the missing letter

"in the 4th second"

02 - Journeys in stages

end of one = start of next

friction: $a = \mu g$

03 - Vertical motion

$a = -9.8$

greatest height: $v = 0$

from above the ground

04 - Bounces and meetings

speed halved

same height, same time

THE FIVE SUVAT EQUATIONS • each one misses one letter

$$v = u + at \text{ (no } s) \quad s = \frac{1}{2}(u + v)t \text{ (no } a) \quad v^2 = u^2 + 2as \text{ (no } t)$$

$$s = ut + \frac{1}{2}at^2 \text{ (no } v) \quad s = vt - \frac{1}{2}at^2 \text{ (no } u)$$

01 The SUVAT Toolkit: One Stage

Constant acceleration in a straight line. Write the five letters in a column, fill in what you know, mark what you want. The letter that is neither given nor wanted tells you which equation to use.

WHEN YOU SEE → YOU DO**“starts from rest” / “comes to rest”**→ $u = 0$ / $v = 0$ **“decelerates at 2 m s^{-2} ”**→ $a = -2$ (motion is positive)**“in the fourth second”**→ $s(4) - s(3)$ — two uses of $s = ut + \frac{1}{2}at^2$ **“passes A ... 20 seconds later passes B”**→ $s = \frac{1}{2}(u + v)t$ when a is unknown **km h^{-1}** → $\div 3.6$ to get m s^{-1} **WORKED MODEL • JUNE 2022**

A motorbike passes P , then Q 10 s later at 28 m s^{-1} . $PQ = 220 \text{ m}$. Find its acceleration.

1 $s = 220$, $v = 28$, $t = 10$, want a — no u

choose the equation

2 $220 = 28(10) - \frac{1}{2}a(100)$

 $s = vt - \frac{1}{2}at^2$

ANSWER $a = 1.2 \text{ m s}^{-2}$

IDEA 01

Practise: The SUVAT Toolkit: One Stage

My questions first, then real past papers.

Q1 DR ESLAM Pick the equation

M1 • TOPIC 4 3 marks

A car accelerates uniformly from 5 m s^{-1} to 25 m s^{-1} over 200 m. Find its acceleration and the time taken.

List s, u, v, a, t ; the one you are not given or asked for tells you the equation.

Q2 DR ESLAM Two intervals

M1 • TOPIC 4 4 marks

A particle with constant acceleration travels 20 m in the first 2 s and 30 m in the next 2 s. Find its initial speed and acceleration.

Use $s = ut + \frac{1}{2}at^2$ for 2 s and for 4 s (50 m).

PAST PAPERS

Q3 PAST PAPER Pearson IAL WME01 · June 2021 · Q2

M1 · TOPIC 4 8 marks

2. A car moves along a straight horizontal road with constant acceleration $a \text{ ms}^{-2}$ where $a > 0$

The car is modelled as a particle.

At time $t = 0$, the car passes point A and is moving with speed $u \text{ ms}^{-1}$

In the first three seconds after passing A the car travels 20 m.

In the fourth second after passing A the car travels 10 m.

The speed of the car as it passes point B is 20 ms^{-1}

Find the time taken for the car to travel from A to B .

(8)

Q4 PAST PAPER Pearson IAL WME01 · June 2022 · Q2

M1 · TOPIC 4 7 marks

2. A motorbike is moving with constant acceleration along a straight horizontal road.

The motorbike passes a point P and 10 seconds later passes a point Q .

The speed of the motorbike as it passes Q is 28 m s^{-1}

Given that $PQ = 220 \text{ m}$,

- (a) find the acceleration of the motorbike,

(3)

- (b) find the distance travelled by the motorbike during the fifth second after passing P

(4)

Q5**PAST PAPER**

Pearson IAL WME01 • January 2024 • Q3

M1 • TOPIC 4 10 marks

3. A van travels with constant acceleration along a straight horizontal road.

The van passes a point A with speed $u \text{ m s}^{-1}$ and 20 seconds later passes a point B with speed 28 m s^{-1}

The distance AB is 400 m.

- (a) Show that $u = 12$ (2)

- (b) Find the time taken for the van to travel from A to the midpoint of AB . (5)

The van has mass 1200 kg.

During its motion the van experiences a constant resistive force of magnitude 260 N

- (c) Find the magnitude of the driving force exerted by the engine of the van as it travels from A to B . (3)

02 Journeys in Stages

When the acceleration changes, start a new SUVAT list. The speed at the end of one stage is the speed at the start of the next. Add the distances.

WHEN YOU SEE → YOU DO

“the acceleration changes instantaneously to ...”

→ new stage, new list

“first model ... second model”

→ solve each model separately and compare

“rough horizontal surface”, coefficient μ

→ deceleration = μg

“maintains this speed”

→ constant stage: distance = speed × time

Stage 1 list

Link speed

Stage 2 list

Add distances / times

WORKED MODEL • JANUARY 2023

PQ = 6x: from u to $2u$ in 12 s. Show $x = 3u$.

$$1 \quad 6x = \frac{1}{2}(u + 2u)(12)$$

$$\text{no } a: s = \frac{1}{2}(u + v)t$$

$$\text{ANSWER} \quad 6x = 18u \Rightarrow x = 3u$$

× **TRAP** Using one list for the whole journey when the acceleration changes halfway.

IDEA 02

Practise: Journeys in Stages

My questions first, then real past papers.

Q6 DR ESLAM Two stages

M1 • TOPIC 4 4 marks

A train starts from rest, accelerates at 0.5 m s^{-2} for 40 s, then decelerates at 1 m s^{-2} until it stops. Find the total distance.

The final speed of stage 1 is the initial speed of stage 2.

Q7 DR ESLAM Friction gives the deceleration

M1 • TOPIC 4 3 marks

A puck slides on rough horizontal ice with $\mu = 0.1$, starting at 6 m s^{-1} . How far does it slide?

PAST PAPERS

Q8 PAST PAPER Pearson IAL WME01 • October 2021 • Q3

M1 • TOPIC 4 10 marks

3. A car is moving at a constant speed of 25 m s^{-1} along a straight horizontal road.

The car is modelled as a particle.

At time $t = 0$, the car is at the point A and the driver sees a road sign 48 m ahead.

Let t seconds be the time that elapses after the car passes A .

In a **first** model, the car is assumed to decelerate uniformly at 6 m s^{-2} from A until the car reaches the road sign.

- (a) Use this first model to find the speed of the car as it reaches the sign.

(2)

The road sign indicates that the speed limit immediately after the sign is 13 m s^{-1} .

In a **second** model, the car is assumed to decelerate uniformly at 6 m s^{-2} from A until it reaches a speed of 13 m s^{-1} . The car then maintains this speed until it reaches the road sign.

- (b) Use this second model to find the value of t at which the car reaches the sign.

(4)

In a **third** model, the car is assumed to move with constant speed 25 m s^{-1} from A until time $t = 0.2$, the car then decelerates uniformly at 6 m s^{-2} until it reaches a speed of 13 m s^{-1} . The car then maintains this speed until it reaches the road sign.

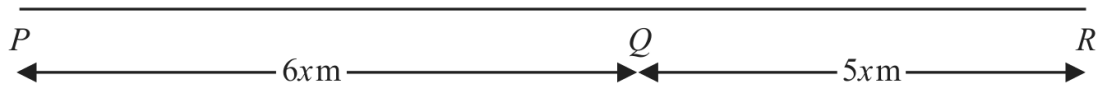
- (c) Use this third model to find the value of t at which the car reaches the sign.

(4)

Q9 PAST PAPER Pearson IAL WME01 · January 2023 · Q5

M1 · TOPIC 4 9 marks

5.

**Figure 2**

Three points P , Q and R are on a horizontal road where PQR is a straight line.

The point Q is between P and R , with $PQ = 6x$ metres and $QR = 5x$ metres, as shown in Figure 2.

A vehicle moves along the road from P to Q with constant acceleration.

The vehicle is modelled as a particle.

At time $t = 0$, the vehicle passes P with speed $u \text{ m s}^{-1}$

At time $t = 12 \text{ s}$, the vehicle passes Q with speed $2u \text{ m s}^{-1}$

Using the model,

(a) show that $x = 3u$ (2)

As the vehicle passes Q , the acceleration of the vehicle changes instantaneously to 1.5 m s^{-2}

The vehicle continues to move with a constant acceleration of 1.5 m s^{-2} and passes R with speed $3u \text{ m s}^{-1}$

Using the model,

(b) find the value of u , (3)

(c) find the distance travelled by the vehicle during the first 14 seconds after passing P (4)

Q10 PAST PAPER Pearson IAL WME01 · January 2026 · Q5

M1 · TOPIC 4 12 marks

5. The points O , A , B and C are on a rough horizontal surface where $OABC$ is a straight line.

A particle is projected horizontally from O and slides across the surface, passing through A , B and C in that order.

The particle passes through

- A with speed $U \text{ m s}^{-1}$
- B with speed $\frac{U}{2} \text{ m s}^{-1}$

The coefficient of friction between the particle and the surface is $\frac{1}{7}$

It takes 0.75 s for the particle to move from A to B .

- (a) Show that $U = 2.1$ to 2 significant figures.

(6)

The particle passes through C with speed $\frac{U}{3} \text{ m s}^{-1}$

- (b) Find the time taken to move from B to C .

(3)

- (c) Find the distance AC .

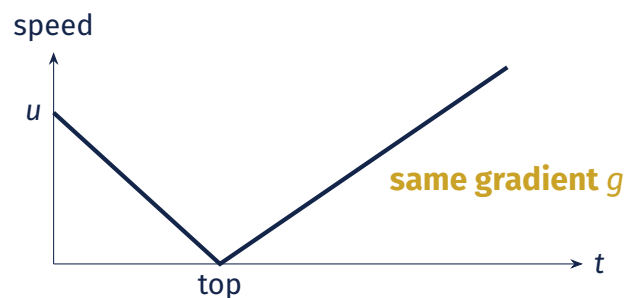
(3)

03 Vertical Motion Under Gravity

Moving freely under gravity means $a = -9.8$ if up is positive. Keep one list for the whole flight — up, top and down — and measure s from the point of projection. Below the start, s is negative.

WHEN YOU SEE → YOU DO

“projected vertically upwards”	→ up +, $a = -9.8$
“greatest height”	→ $v = 0$ at the top
“from a point 24.5 m above the ground”	→ the ground is at $s = -24.5$
“the time for which it is above C”	→ two roots of one quadratic, subtract
“sketch a speed–time graph”	→ V shape: down to 0 at the top, up again
“speed less than 14.7”	→ find where $v = \pm 14.7$



WORKED MODEL • JANUARY 2021

A stone is projected upwards at 20 m s^{-1} from O , 5 m above the ground. Find its speed when it is 2 m above the ground.

1 $s = -3$ (3 m below O), $u = 20$, $a = -9.8$

list with signs

2 $v^2 = 400 + 2(-9.8)(-3) = 458.8$

$v^2 = u^2 + 2as$

ANSWER $v = 21.4 \text{ m s}^{-1}$

IDEA 03

Practise: Vertical Motion Under Gravity

My questions first, then real past papers.

Q11 **DR ESLAM** Up and back down

M1 • TOPIC 4 3 marks

A ball is thrown vertically upwards at 14.7 m s^{-1} . Find the greatest height and the time until it returns to the point of projection.

Q12 **DR ESLAM** From above the ground

M1 • TOPIC 4 4 marks

A stone is thrown vertically upwards at 10 m s^{-1} from a point 20 m above the ground. Find the time until it hits the ground.

Up is +: the ground is at $s = -20$. One equation, then the positive root.

PAST PAPERS

Q13 PAST PAPER Pearson IAL WME01 · January 2021 · Q1

M1 · TOPIC 4 6 marks

1. A small stone is projected vertically upwards with speed 20 m s^{-1} from a point O which is 5 m above horizontal ground. The stone is modelled as a particle moving freely under gravity.

Find

- (a) the speed of the stone at the instant when it is 2 m above the ground, (2)
- (b) the total time between the instant when the stone is projected from O and the instant when it first strikes the ground. (4)

Q14 PAST PAPER Pearson IAL WME01 · January 2022 · Q4

M1 · TOPIC 4 8 marks

4. At time $t = 0$, a small ball is projected vertically upwards from a point A which is 24.5 m above the ground. The ball first comes to instantaneous rest at the point B , where $AB = 19.6$ m and first hits the ground at time $t = T$ seconds.

The ball is modelled as a particle moving freely under gravity.

- (a) Find the value of T .

(6)

- (b) Sketch a speed-time graph for the motion of the ball from $t = 0$ to $t = T$ seconds.

(No further calculations are needed in order to draw this sketch.)

(2)

Q15**PAST PAPER**

Pearson IAL WME01 · October 2022 · Q5

M1 · TOPIC 4 9 marks

5. A small ball is projected vertically upwards with speed 29.4 m s^{-1} from a point A which is 19.6 m above horizontal ground.

The ball is modelled as a particle moving freely under gravity until it hits the ground. It is assumed that the ball does not rebound.

- (a) Find the distance travelled by the ball while its speed is less than 14.7 m s^{-1} (3)
- (b) Find the time for which the ball is moving with a speed of more than 29.4 m s^{-1} (3)
- (c) Sketch a speed-time graph for the motion of the ball from the instant when it is projected from A to the instant when it hits the ground. Show clearly where your graph meets the axes. (3)

Q16 PAST PAPER Pearson IAL WME01 · January 2024 · Q6

M1 · TOPIC 4 12 marks

6. A particle is projected vertically upwards from a point A with speed 24 m s^{-1}

The point A is 2.5 m vertically above the point B .

Point B lies on horizontal ground.

The particle moves freely under gravity until it hits the ground at B with speed $V \text{ m s}^{-1}$

After hitting the ground the particle does not rebound.

- (a) Find the value of V . (3)

- (b) Find the time taken for the particle to reach B . (3)

The point C is 10 m vertically above A .

- (c) Find the length of time for which the particle is above C . (4)

- (d) Sketch a speed-time graph for the motion of the particle from projection to the instant that it reaches B . (No further calculations are required.) (2)

Q17 PAST PAPER Pearson IAL WME01 • June 2024 • Q5

M1 • TOPIC 4 14 marks

5. A parachute is used to deliver a box of supplies. The parachute is attached to the box.
- the parachute and box are dropped from rest from a helicopter that is hovering at a height of 520 m above the ground
 - the parachute and box fall vertically and freely under gravity for 5 seconds, then the parachute opens
 - from the instant the parachute opens, it provides a resistance to motion of magnitude 3200 N
 - the parachute and box continue to fall vertically downwards after the parachute opens
 - the parachute and box are modelled throughout the motion as a particle P of mass 250 kg
- (a) Find the distance fallen by P in the first 5 seconds. (2)
- (b) Find the speed with which P lands on the ground. (7)
- (c) Find the total time from the instant when P is dropped from the helicopter to the instant when P lands on the ground. (3)
- (d) Sketch a speed-time graph for the motion of P from the instant when P is dropped from the helicopter to the instant when P lands on the ground. (2)

Q18 PAST PAPER Pearson IAL WME01 • June 2025 • Q3

M1 • TOPIC 4 8 marks

3. A particle is projected vertically upwards with speed $U \text{ m s}^{-1}$ from a point A .

The point A is 12 m vertically above the point B .

Point B is on horizontal ground.

The particle moves freely under gravity until it hits the ground at B .

The time taken for the particle to travel from A to B is 4 seconds.

- (a) Find the value of U . (3)
- (b) Find the speed of the particle as it hits the ground at B . (3)
- (c) Sketch a speed-time graph for the motion of the particle from the instant it leaves A to the instant it reaches B . (No further calculations are required.) (2)

04 Bounces and Two Particles Meeting

A bounce starts a new flight: find the speed at impact, change it (halved, or given), and start again with u upwards. Two particles meet when they are at the same height at the same time: write each height in t and set them equal.

WHEN YOU SEE → YOU DO

“the speed is instantaneously reduced by 50%”

→ new $u = \frac{1}{2}$ the impact speed

“rebounds to a height of 2 m”

→ $0 = u^2 - 2g(2)$ gives the rebound speed

“impulse received on impact”

→ $m(v + u)$ — the directions reverse

“projected 1 second later”

→ its time is $t - 1$

“collide” / “at the same height”

→ equal heights from one reference level

WORKED MODEL • OCTOBER 2019

Released from 40 m. Each bounce halves the speed. Height after the third bounce?

1 impact speed $\sqrt{2(9.8)(40)} = 28$

$$v^2 = 2gs$$

2 after 3 bounces: $28 \rightarrow 14 \rightarrow 7 \rightarrow 3.5$

halve each time

ANSWER $h = \frac{3.5^2}{19.6} = 0.625 \text{ m}$

IDEA 04

Practise: Bounces and Two Particles Meeting

My questions first, then real past papers.

Q19**DR ESLAM**

One bounce

M1 • TOPIC 4**3 marks**

A ball is dropped from 10 m. It rebounds with half the speed it hits the ground with. Find the greatest height after the bounce.

Q20**DR ESLAM**

Two particles meet

M1 • TOPIC 4**4 marks**

P is dropped from 40 m. At the same instant Q is thrown upwards from the ground below at 20 m s^{-1} . When and at what height do they meet?

Heights equal: $40 - 4.9t^2 = 20t - 4.9t^2$.

PAST PAPERS

Q21 PAST PAPER Pearson IAL WME01 · October 2019 · Q2

M1 · TOPIC 4 9 marks

2. A small ball is released from rest from a point that is 40 m above horizontal ground. The ball bounces on the ground and rebounds vertically. Each time the ball bounces on the ground, the speed of the ball is instantaneously reduced by 50%. The ball is modelled as a particle moving freely under gravity, from the instant when it is released until it first hits the ground, and between each successive bounce.
- (a) Find the time from the instant when the ball is released from rest to the instant when it hits the ground for the second time. (5)
- (b) Find the maximum height reached by the ball above the ground after the ball's third bounce. (4)

Q22**PAST PAPER**

Pearson IAL WME01 · January 2020 · Q3

M1 · TOPIC 4**11 marks**

3. A particle, P , is projected vertically upwards with speed U from a fixed point O . At the instant when P reaches its greatest height H above O , a second particle, Q , is projected with speed $\frac{1}{2}U$ vertically upwards from O .
- (a) Find H in terms of U and g . (2)
- (b) Find, in terms of U and g , the time between the instant when Q is projected and the instant when the two particles collide. (6)
- (c) Find where the two particles collide. (3)

Q23**PAST PAPER**

Pearson IAL WME01 · October 2024 · Q7

M1 · TOPIC 4**11 marks**

- 7 At time $t = 0$, a small ball A is projected vertically upwards with speed 8 m s^{-1} from a fixed point on horizontal ground.

The ball hits the ground again for the first time at time $t = T_1$ seconds.

Ball A is modelled as a particle moving freely under gravity.

- (a) Show that $T_1 = 1.63$ to 3 significant figures.

(2)

After the first impact with the ground, A rebounds to a height of 2 m above the ground.

Given that the mass of A is 0.1 kg,

- (b) find the magnitude of the impulse received by A as a result of its first impact with the ground.

(5)

At time $t = 1$ second, another small ball B is projected vertically upwards from another point on the ground with speed 5 m s^{-1}

Ball B is modelled as a particle moving freely under gravity.

At time $t = T_2$ seconds ($T_2 > 1$), A and B are at the same height above the ground for the first time.

- (c) Find the value of T_2

(4)

Q24 PAST PAPER Pearson IAL WME01 • October 2025 • Q4

M1 • TOPIC 4 9 marks

4. The point A is 10 m above horizontal ground.

At time $t = 0$, a particle P is projected vertically upwards with speed 5 m s^{-1} from A . Particle P moves freely under gravity.

- (a) Find the greatest height above A reached by P .

(3)

The point B is on the ground, vertically below A .

At time $t = 1$ second, a particle Q is projected vertically upwards with speed 7 m s^{-1} from B .

Particle Q moves freely under gravity.

Particles P and Q collide at time $t = T$ seconds.

- (b) Find the value of T .

(4)

- (c) Find the speed of P at the instant immediately before the particles collide.

(2)

Q25**PAST PAPER**

Pearson IAL WME01 • June 2026 • Q4

M1 • TOPIC 4 9 marks

4. The point O is 28.4 m above a horizontal plane. A small ball is projected vertically upwards from O with speed $U \text{ m s}^{-1}$ and first hits the plane with speed 24 m s^{-1}

Air resistance is assumed to be negligible.

- (a) Find the value of U .

(3)

Each time the ball hits the plane, the speed of the ball is instantaneously halved.

- (b) Find the **total** time, **after the ball has hit the plane for the first time**, for which the ball is at least 4 m above the plane.

(6)

(Total for Question 4 is 9 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|--------------------------|---|
| 01 | SUVAT | List s, u, v, a, t ; the missing letter picks the equation. |
| 02 | Stages | New list each stage; end speed = next start speed. |
| 03 | Vertical | Up +, $a = -9.8$; below the start $s < 0$. |
| 04 | Bounces, meetings | New flight after a bounce; equal heights, same t . |

BOTTOM LINE Positive direction first · five letters · one equation · check the sign of every answer.

Answers

Constant Acceleration in 1D · for the teacher. Every answer checked with a computer algebra system.

IDEA 01 · THE SUVAT TOOLKIT: ONE STAGE

- Q1** Dr Eslam · Pick the equation $625 = 25 + 400a \Rightarrow a = 1.5 \text{ m s}^{-2}; t = \frac{40}{3} \approx 13.3 \text{ s}$
- Q2** Dr Eslam · Two intervals $20 = 2u + 2a, 50 = 4u + 8a: a = 2.5, u = 7.5$
- Q3** Pearson IAL WME01 · June 2021 · Q2 $t = 9.5 \text{ s}$
- Q4** Pearson IAL WME01 · June 2022 · Q2 (a) 1.2 m s^{-2} (b) 21.4 m
- Q5** Pearson IAL WME01 · January 2024 · Q3 (a) $400 = \frac{1}{2}(u + 28)(20) \Rightarrow u = 12$ (b) $5\sqrt{29} - 15 \approx 11.9 \text{ s}$ (c) 1220 N

IDEA 02 · JOURNEYS IN STAGES

- Q6** Dr Eslam · Two stages $v = 20; 400 + 200 = 600 \text{ m}$
- Q7** Dr Eslam · Friction gives the deceleration $a = -0.1g = -0.98; 0 = 36 - 1.96s \Rightarrow s = 18.4 \text{ m}$
- Q8** Pearson IAL WME01 · October 2021 · Q3 (a) 7 m s^{-1} (b) $t = \frac{36}{13} \approx 2.77 \text{ s}$ (c) $t = \frac{168}{65} \approx 2.58 \text{ s}$
- Q9** Pearson IAL WME01 · January 2023 · Q5 (a) $6x = \frac{1}{2}(u + 2u)(12) \Rightarrow x = 3u$ (b) $u = 9$ (c) 201 m
- Q10** Pearson IAL WME01 · January 2026 · Q5 (a) $a = \frac{g}{7} = 1.4; \frac{u}{2} = U - 1.4(0.75) \Rightarrow U \approx 2.1$ (b) 0.25 s (c) $AC = 1.4 \text{ m}$

IDEA 03 · VERTICAL MOTION UNDER GRAVITY

- Q11** Dr Eslam • Up and back down
 $H = \frac{14.7^2}{19.6} = 11.0 \text{ m}; t = 3 \text{ s}$
- Q12** Dr Eslam • From above the ground
 $-20 = 10t - 4.9t^2 \Rightarrow t = 3.28 \text{ s}$
- Q13** Pearson IAL WME01 • January 2021 • Q1
 (a) 21.4 m s^{-1} (b) 4.32 s
- Q14** Pearson IAL WME01 • January 2022 • Q4
 (a) $T = 5$ (b) V shape: down to 0 at $t = 2$, then up to 29.4 at $t = 5$
- Q15** Pearson IAL WME01 • October 2022 • Q5
 (a) $22.05 \approx 22.1 \text{ m}$ (b) 0.606 s (c) V shape from 29.4 to 0 at $t = 3$, up to the landing speed
- Q16** Pearson IAL WME01 • January 2024 • Q6
 (a) $V = 25$ (b) 5 s (c) 3.98 s (d) V shape: 24 down to 0 , up to 25 at $t = 5$
- Q17** Pearson IAL WME01 • June 2024 • Q5
 (a) $122.5 \approx 123 \text{ m}$ (b) 4 m s^{-1} (c) 20 s (d) $(0, 0) \rightarrow (5, 49) \rightarrow (20, 4)$
- Q18** Pearson IAL WME01 • June 2025 • Q3
 (a) $U \approx 16.6$ (b) 22.6 m s^{-1} (c) V shape through $(0, 16.6)$, $(1.69, 0)$, $(4, 22.6)$

IDEA 04 • BOUNCES AND TWO PARTICLES MEETING

- Q19** Dr Eslam • One bounce hits at 14 m s^{-1} , leaves at 7: $\frac{49}{19.6} = 2.5 \text{ m}$
- Q20** Dr Eslam • Two particles meet $t = 2 \text{ s}$, height 20.4 m
- Q21** Pearson IAL WME01 • October 2019 • Q2 (a) $\frac{40}{7} \approx 5.71 \text{ s}$ (b) 0.625 m
- Q22** Pearson IAL WME01 • January 2020 • Q3 (a) $H = \frac{U^2}{2g}$ (b) $\frac{U}{g}$ (c) at O
- Q23** Pearson IAL WME01 • October 2024 • Q7 (a) $T_1 = \frac{16}{g} \approx 1.63$ (b) $0.1(8 + \sqrt{4g}) \approx 1.43 \text{ N s}$
(c) $T_2 = \frac{99}{68} \approx 1.46$
- Q24** Pearson IAL WME01 • October 2025 • Q4 (a) $\frac{25}{19.6} \approx 1.28 \text{ m}$ (b) $T \approx 1.86$ (c) 13.2 m s^{-1}
- Q25** Pearson IAL WME01 • June 2026 • Q4 (a) $U = 4.4$ (b) 1.65 s